

JOSHUA ONUNGWE, M.Sc.

Senior Software Engineer (Frontend)

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SUMMARY

I am a Staff-level Software Engineer with over 13 years of experience architecting and delivering highly scalable, data-intensive applications. I am passionate about solving technical problems and user-centric design. Over the years, I have used my expertise in the frontend, backend and distributed systems to build products that are technically sound, intuitive and impactful. I am an advocate for continuous learning. I love crafting and evolving design systems, and most recently, I have been experimenting with AI engineering, workflows and techniques. With an M.Sc. in Data Science, I have also been working on transforming complex data into meaningful user experiences and driving innovation through custom LLMs, AI and machine learning models.

TECHNICAL PROFICIENCIES

Frontend: JavaScript (ES6+), TypeScript, React, React Native, Redux (Toolkit), Next.js, Cypress, Jest, React Testing Library, Storybook, Design Systems, Module Federation, Micro-frontends, AXE-Core, ARIA

Backend: Node.js, Python, Elixir (Phoenix), Golang, Java, REST/GraphQL, Microservices, Serverless (AWS Lambda), PostgreSQL, MongoDB, Redis, Kafka

DevOps & Tools: Docker, Kubernetes, AWS (Lambda, EC2, RDS, S3), CI/CD (CircleCI, GitHub Actions, Jenkins), Git, Terraform, Docusaurus, Browserstack

Other: Accessibility (WCAG compliance), AI Agents, Machine Learning (NLP, Graph Mining), Blockchain (EVMs), Agile/Scrum, TDD/BDD, ANTLR

WORK EXPERIENCE

Senior Software Engineer (Frontend), Kobai Inc - 04/2022 - 05/2025 - Ontario, Canada

- Architected an enterprise knowledge graph platform in React and TypeScript, managing over 500 million records and visualizing petabyte-scale datasets for Fortune 500 clients.
- Implemented Module Federation across a multi-repository micro-frontend landscape, enabling independent deployment of core services and reducing cross-team build dependencies.
- Engineered a shared, ANTLR-based language parser for universal validation across the frontend and backend, replacing the legacy editor with a high-performance Monaco-based IDE. This eliminated 100% of freezing issues and drove a 40% increase in feature adoption.
- Led optimizations of our Docker-based build system on CircleCI, cutting pipeline times by 50% and saving an estimated \$100,000 annually in operational costs.
- Governed the evolution of the company-wide design system using Storybook and React, improving designer-developer collaboration, documentation and accelerating feature development by an estimated 30%.
- Led a team of frontend developers, enforcing TDD/BDD practices with Jest and Playwright, and achieving WCAG accessibility compliance.
- Partnered with DevOps to re-architect the data ingestion pipeline, implementing a direct-to-Databricks streaming solution that reduced processing times for large datasets by over 95% (from hours to minutes).

Lead Frontend Engineer, Aya Payments Inc. - 03/2019 - 04/2022- Ontario, Canada

- Re-engineered a failing claims app in 8 weeks with React Native and Redux Toolkit, replacing manual workflows and driving \$3.5 million in revenue within the first year.
- Implemented code-splitting and smart caching, cutting page load 40 % while supporting 3× traffic growth.
- Managed and mentored a team of 13 developers, fostering a culture of high-quality code through rigorous reviews, introducing QA best practices, and implementing a reusable Storybook component library on an AWS serverless stack (Lambda, DynamoDB).

- Integrated secure payment workflows with external fintech stakeholders via REST APIs, ensuring reliable and compliant transaction processing.

Senior Frontend Developer, Softcom Limited - 09/2018 - 01/2019 - Lagos, Nigeria

- Engineered a microservice architecture for NPower using Node.js and Docker on Kubernetes, achieving 99.9% uptime for 12 million users.
- Developed high-throughput Node.js microservices using Redis and RabbitMQ to support 1.5 million marketplace participants.
- Mentored junior and mid-level developers on advanced React patterns and best practices, improving team-wide code quality and consistency.
- Built an advanced React Form Builder suite (databeaver), improving stability by 25% and speeding up client deployments.

PERSONAL PROJECTS

DiscoverNow (<https://discovernow.xyz>)

- Architected and led the end-to-end development of a cross-platform educational application using Next.js and React Native, aimed at cultural preservation.
- Integrated multiple LLMs to create dynamic, personalized learning paths, served via a scalable backend on AWS with a PostgreSQL database.
- Engineered a fully automated CI/CD pipeline with GitHub Actions for rapid and reliable feature deployment.

AiutaConnect (<https://aiutaconnect.com>)

- Engineered a secure, accessible social platform in Next.js and React Native for a global community of seniors who belong to the University of the Third Age.
- Implemented a real-time communication infrastructure using WebSockets and Node.js to enable private and group messaging.
- Designed and delivered a seamless online course module, allowing universities to offer content directly to users.

EDUCATION AND CERTIFICATES

M.Sc. in Data Science and Analytics, Ryerson University (Toronto Metropolitan University)

B.Sc. in Information and Communication Science, University of Ilorin